



## Socio-economic features of sea cucumber fisheries in southern coast of Kenya

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### ABSTRACT

A socio-economic assessment was conducted at Vanga, Shimoni, Majoreni and Gazi villages in the Kenyan south coast with focus on the sea cucumber fishing patterns, the social and economic characteristics of the fisher communities, the contribution of sea cucumbers to the local livelihoods, and analysis of the management systems. The results indicate that sea cucumber fishers are mainly men. Fishing is done in sub-tidal areas (3–10 m deep) and inter-tidal areas depending on the species being targeted. Those who fish in the sub-tidal areas do skin diving without using SCUBA diving gear. Sea cucumber fishing is heavily done during the northeast monsoon season when the sea is calm and water is clear. About 32% of the sea cucumber fishers also collect other marine products such as octopus. The sea cucumbers are sold fresh from the sea to local first level middlemen who process and sell them to the second level middlemen and exporters in Mombasa. The fishers occasionally borrow money from first level middlemen especially when they fail to catch sea cucumbers but this in turn creates conditions of dependence and possible exploitation. Almost all sea cucumber fishers have stated that they are not willing to make sea cucumbers part of their daily diet. The economic value of the product was substantial; the average monthly revenue for dry sea cucumbers in the area was estimated to US\$ 8000. The relative highest profits are derived from juvenile species, thus there is an economic incentive hindering local stocks to reach sexual maturity, which in turn may create a situation in which recruitment success is highly dependent on faraway populations. The present management system falls into general fisheries regulations and was found weak. No specific management plan for sea cucumbers was found.

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### 1. Introduction

The sea cucumber fishery is an important source of livelihood to many households in the coast of Kenya [1]. It has existed since the 1900s and the number of fishers involved has been increasing over time. In the last 50 years the rate at which sea cucumbers are harvested has increased almost sevenfold reaching 28,376 tones (dry weight) in 2003 [2]. As fishing effort has increased over time due to the need to meet the high demand for sea cucumber products from the Asian market, reports of over-exploitation are also increasing [3]. It is important to note that fish production (including harvesting of sea cucumbers) from the inshore areas is determined by among other factors, the amount and quality of effort that is applied in the fishery and the availability of stocks [4]. Over the

years the number of people joining the sea cucumber fishery has increased thereby putting considerable pressure on the limited stocks. Catches have been declining over the last 10 years and fishers are catching smaller and reproductively immature individuals of at least two commercial species (*Holothuria fuscogilva* and *Holothuria scabra*) at the main landing sites in Kenya [5].

In fact some of the high value species such as *H. fuscogilva*, which is locally known as 'pauni mweupe', have become extremely scarce and the previously low-value species that were initially not harvested are now being harvested. Although sea cucumbers are not eaten by local people the sea cucumber fishery has become an important income source.

Sea cucumber catches have declined by over half from about 62,000 tones to 28,376 tones (dry weight) between 1984 and 2004 despite having a peak in 1992 [6,7]. The 1992 peak was only associated with the introduction of SCUBA diving techniques to harvest sea cucumbers which has rendered deep water stocks, previously inaccessible to the local fishers, vulnerable to exploitation. By 2005, the price of sea cucumbers had increased from an average of KSh.51/kg to KSh.312/kg (US\$ 1 = KSh.76) [6,7]. This six-fold

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increase in price has occurred due to increased demand for sea cucumber products in the international market and global supply has not increased at the same pace thus resulting in a situation where demand exceeds supply [7,8].

As sea cucumber catches decline, the question that it raises is, what are the implications of over-exploitation of sea cucumbers on the welfare of the fishers in Kenya? For a long time, the availability of sea cucumber stocks has been taken for granted and this has resulted in non-responsive management of the fishery. In addition, the fisheries department has given focus to finfish fisheries, habitat degradation, destructive dynamite fishing techniques and management arrangements in the form of Marine Protected Areas (MPAs) that include marine parks and reserves. According to Beadle [7], Kenya's sea cucumber fishery may be on the verge of becoming economically unviable and current regulations are inefficient to ensure its future sustainability. Consequently, there is need for additional conservation and management measures to be put in place, and managers and policy makers should address the underlying factors which influence the effectiveness of management regimes. Like other marine resources, the complexity of managing sea cucumber resources cannot be overstated. It has been observed that the management of marine resources is highly complex and characterized, among other things, by extremely high levels of uncertainty [9–12] and complex governance issues e.g. Jentoft [13]. The inadequacy of information about the state and exploitation of sea cucumber and other marine resources has often led managers to make management decisions and formulate policies that are not responsive to the emerging challenges.

The main objectives of this study are (1) to determine the sea cucumber fishing patterns, the social and economic characteristics of fisher communities and the contribution of sea cucumbers to the local livelihoods, and (2) to assess marine resource management systems to provide a baseline for planning, management and future improvements.

## 2. Research approach

### 2.1. Study sites

The study was conducted in four sites in the Kenyan south coast namely Vanga, Majoreni, Shimoni and Gazi (Fig. 1). These are important sea cucumber fishing villages where sea cucumbers are landed on a daily basis and regular records are maintained [14,22]. In 2003, these sites (i.e. the former Kwale district) produced more than 80% of the sea cucumber catch in Kenya. Each of the sites has its own unique characteristics. Vanga is a typical fishing village with high levels of sea cucumber harvesting. It is located in the southern most part of the Kenyan coast at the border of Kenya and Tanzania. It also has the influence of migrant fishers due to its fairly rich fishing ground. Majoreni is a rural village that depends on both fishing and peasant farming. It is currently experiencing declining fish catches due to overfishing and use of destructive fishing techniques thus many fishers from the village are getting involved in sea cucumber fishing. Shimoni is a rural settlement adjacent to an MPA (Kisite-Mpunguti Marine Park and Reserve). It consists of three distinct fishing villages (Shimoni, Kibuyuni and Kiromo) where sea cucumber fishing is an important activity and data was collected from all the three villages. Gazi is a rural fishing village located on the shores of Gazi Bay about 60 km south of Mombasa city. It attracts migrant fishers that engage in different fisheries including the sea cucumber fishery. Data was also collected from the major sea cucumber dealers (second level middlemen) and exporters in Mombasa city.

### 2.2. Methods

Primary data was collected using a combination of four techniques namely questionnaires, semi-structured interviews, key-informant interviews, and focus group discussions, as proposed by Bunce et al. [15] and de la Torre-Castro et al. [14]. A summary of the

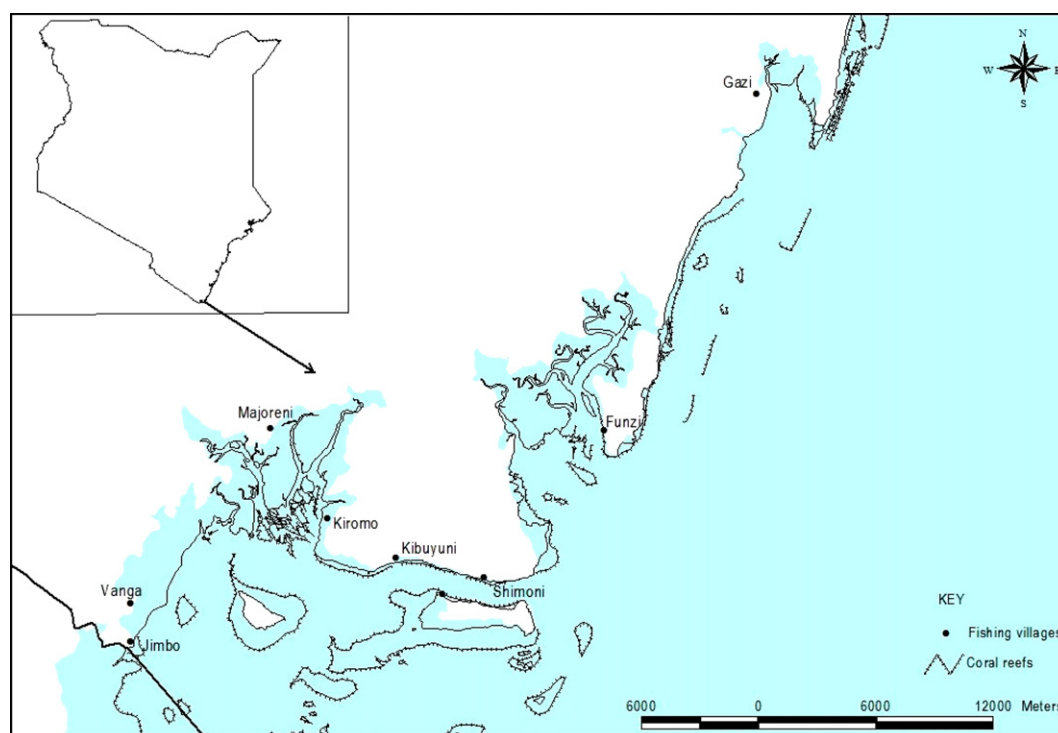


Fig. 1. Southern coast of Kenya showing the sea cucumber fishing villages integrated in this study, i.e. Vanga, Majoreni, Shimoni and Gazi.

application of these techniques is presented below. Three categories of respondents were identified before primary data collection began namely sea cucumber fishers, first level middlemen who are based at specific sea cucumber landing beaches, and second level middlemen and exporters who are based in the major town of Mombasa.

1. **Questionnaires:** A structured questionnaire was administered to a selected sample of sea cucumber fishers to obtain quantitative data that could be statistically analyzed. Sixty respondents were selected randomly from among the sea cucumber fishers. A separate questionnaire was administered to the middlemen and exporters. Since the middlemen and exporters were found to be few, all of them were surveyed. The questionnaire survey was conducted once towards the end of the data collection after the research team had established a rapport with the fishers and middlemen.
2. **Semi-structured interviews:** Respondents were randomly selected and included 62 sea cucumber fishers, eight middlemen and two exporters. Semi-structured interviews were conducted using interview guides/semi-structured questionnaires with open-ended questions. Some respondents were interviewed in their homes while others were interviewed at the beach where they landed their catch or conduct their businesses. Using this method, it was possible to probe for answers, follow-up the original questions and pursue new lines of questions. It created room for two-way interactions and exchange of information between the interviewer and the respondent. At the end of each day, the research team sat together to review the results.
3. **Key-informant interviews:** A key-informant interview technique was used to gather information from the opinion leaders. Thirteen key informants were interviewed. The snowball method (in which information is gathered by asking an initial informant to suggest other informants) was used to identify key informants in the villages. These key informants are people who hold some respected positions in the society. The key informants provided insights on many issues that needed further clarifications and helped in the validation of information collected using the other research methods.
4. **Focus group discussions:** Seven focus group discussions were conducted: two at Vanga and at Majoreni and three at Shimoni. Each focal group consisted of 5–10 respondents. A set of open-ended questions were used to prompt participants into free discussions focusing on the issues under the study. The focus group interviews took place on Friday when the fishers did not go fishing since this is their prayer day.

The above data collection methods were applied to the three categories of respondents as shown in Table 1 below.

The data of sea cucumber landings at Vanga, Shimoni, Majoreni and Kwale district was obtained from records from the Fisheries Department. The data series is shown from 2004 or 2005 to 2008. The data was checked first to see if the quality was good enough for the analysis, by comparing data sets with results from our

interviews, and talking with the technical assistants to acquire more information.

### 3. Results

#### 3.1. Socio-demographic characteristics of the sea cucumber fishers

About 80% of the interviewees are household heads while 20% still live with their parents and only make some extra contributions to their household income. In this study, a household is defined as a social and economic unit where people live, cook and eat from a common pot and share resources and tasks. The average household size in the four fishing villages is seven persons, with households in Vanga village having relatively more members per household than those in Majoreni, Shimoni and Gazi villages. Furthermore, 88.3% of the sea cucumber fishers are aged 19–40 years while 6.7% are aged 41–50 years and 5.0% are aged 51–60 years. This implies that if we assume that the most economically active age falls between 18 years and 55 years then we can conclude that 95% of the sea cucumber fishers are in this productive age category. None of the sea cucumber fishers is over 60 years old. This is completely different from the finfish fishery where even elderly people who are over 60 years old are still actively fishing. The study also showed that 73.3% of the fishers are married, 16.7% are single while 10% fall in other categories including those that are separated, divorced, and those whose spouses had died. Furthermore, it was established that 21.1% of the sea cucumber fishers have no formal education, 66.7% have reached various levels in primary school, 8.3% have reached different levels of secondary school and only 3.9% have post secondary education. The respondents in Shimoni and Gazi villages have attended school for more years (5.7–6.5 years) than those from Vanga and Majoreni villages (3–5.4 years). The general social, demographical and educational characteristics of the villages are shown in Tables 2 and 3 below.

#### 3.2. Fishing activities

The study established that 68% of the sea cucumber fishers are engaged full-time in sea cucumber collection. The other 32% of the sea cucumber fishers are only engaged in it part-time since they collect sea cucumber as bycatch while they target finfish, octopus and lobsters. Some of the part-time fishers are also engaged in subsistence farming while some are engaged in casual jobs to supplement their incomes. Sea cucumber fishing mainly takes place during the northeast monsoon (NEM) which occurs between September and April. During NEM the sea is calm and the water is clear thus making it easy to spot and collect sea cucumbers. During

**Table 2**

General and socio-demographic characteristics of sea cucumber fishers in the four villages of the study in southern Kenya.

| Characteristic                                     | Village |         |          |       |       |
|----------------------------------------------------|---------|---------|----------|-------|-------|
|                                                    | Gazi    | Shimoni | Majoreni | Vanga | Total |
| Sex (%)                                            |         |         |          |       |       |
| Female                                             | 0       | 0       | 0        | 0     | 0     |
| Male                                               | 100     | 100     | 100      | 100   | 100   |
| Educational attainment (no. of years of schooling) | 6.5     | 5.7     | 5.4      | 3.0   | 4.9   |
| Marital status (%)                                 |         |         |          |       |       |
| Married                                            | 66.7    | 73.0    | 90.9     | 61.9  | 71.7  |
| Never married                                      | 0.0     | 24.0    | 9.1      | 14.3  | 16.7  |
| Other                                              | 33.3    | 3.0     | 0.0      | 23.8  | 11.7  |
| Household size                                     | 5.3     | 6.6     | 7.45     | 7.6   | 7.25  |

**Table 1**

The identified categories of respondents in the sea cucumber fishery and the research methods applied.

| Method                     | Fishers | Middlemen | Management authorities |
|----------------------------|---------|-----------|------------------------|
| Questionnaires             | Yes     | Yes       | No                     |
| Semi-structured interviews | Yes     | Yes       | Yes                    |
| Key-informant interviews   | Yes     | Yes       | No                     |
| Focus group discussions    | Yes     | No        | No                     |

**Table 3**

School attendance among fishermen in the four villages of the study in southern Kenya.

|           |           | Gazi  | Majoreni | Shimoni | Vanga | Grand mean |
|-----------|-----------|-------|----------|---------|-------|------------|
| Education | None      | 22.7% | 9.1%     | 19.2%   | 33.3% | 21.1%      |
|           | Primary   | 58.0% | 81.8%    | 60.3%   | 66.7% | 66.7%      |
|           | Secondary | 19.3% | 0%       | 13.9%   | 0%    | 8.3%       |
|           | Other     | 0%    | 9.1%     | 6.6%    | 0%    | 3.9%       |

the southeast monsoon (SEM) that occurs between May and August, sea cucumber fishers get involved in other activities that could earn them income since the sea gets very rough and the water gets turbid during this time thereby making it difficult for fishers to spot the sea cucumbers. The latter suggests that an involuntary “closed season” exists due to climatic conditions.

The full-time sea cucumber fishers conduct their fishing operations for six days per week. Since about 90% of the fishers are Muslims, they normally do not go fishing on Friday which is their special prayer day. In each month fishing is heavily done during spring tide which is a period locally known as ‘*bamvua*’. The fishers prefer to rest during neap tide which is a period locally known as ‘*kimbuya*’. Most part-time fishers go to the sea for two to four days per week. Most of the sea cucumber fishers spend an average of 3 h a day doing actual fishing. In addition, they spend 1 h traveling from the beach to the fishing ground and another 1 h traveling back to the landing beach.

In this study, fishing effort has been measured by the pieces of sea cucumbers landed per person (‘fisherman’) per day. The pieces of sea cucumbers landed by each person is influenced by a number of factors including the type of fishing vessel used, time spent fishing, the diving and/or snorkeling equipment and skills (experience in sea cucumber fishing), the tidal regime and condition of the fishing ground. Sea cucumber fishing does not require complicated fishing gear since they are picked by hand from their habitats. On average, 16 pieces of sea cucumbers are landed per person per day by full-time sea cucumber fishers. The average sea cucumber catches at the main landing beaches range from eight pieces/person/day at Vanga, 17 pieces/person/day at Shimoni, 21 pieces/person/day at Majoreni, and six pieces/person/day at Gazi. It is evident from these figures that the pieces of sea cucumbers landed at Gazi and Shimoni by each fisher per day have significantly decreased compared to the estimates by Muthiga and Ndirangu [5] of 12–30 pieces/man/day at Gazi and 20–24 pieces/man/day at Shimoni. Table 4 shows the actual effort in the different villages.

The average sea cucumber catches at Shimoni are raised by the relatively higher catches landed by fishers from Kiromo village. The higher catches may be attributed to the higher abundance because of closeness of the fishing ground to the MPA (Kisite-Mpunguti Marine Park and Reserve).

From interview results, it was established that *H. scabra* (*mchanga*) is the most commonly landed commercial sea cucumber species in all sites except Gazi (Table 5). The sea cucumber species

**Table 4**

Fishery effort expressed as the number of pieces of sea cucumbers landed per person (fisherman) per day in the four villages of the study in southern Kenya.

| Fishing villages | N  | Average pieces of sea cucumbers landed per day | Std. error |
|------------------|----|------------------------------------------------|------------|
| Vanga            | 8  | 8                                              | 0.923      |
| Majoreni         | 21 | 21                                             | 2.439      |
| Shimoni          | 17 | 17                                             | 2.428      |
| Gazi             | 6  | 6                                              | 2.082      |
| Total            | 60 | 14                                             | 1.328      |

**Table 5**

Commercial sea cucumber species and their associated contribution to the landings in the four villages of the study in southern Kenya.

| Species                      | Percentage contribution of each of the sea cucumber species to catches in each village |          |         |                      |                    |      |
|------------------------------|----------------------------------------------------------------------------------------|----------|---------|----------------------|--------------------|------|
|                              | Vanga                                                                                  | Majoreni | Shimoni | Shimoni 2 (Kibuyuni) | Shimoni 3 (Kiromo) | Gazi |
| <i>Holothuria scabra</i>     | 43                                                                                     | 95       | 20      | 22                   | 23                 | 2    |
| <i>H. nobilis</i>            | 3                                                                                      | –        | 17      | 8                    | 8                  | 13   |
| <i>H. fuscogilva</i>         | 8                                                                                      | –        | 14      | 8                    | 4                  | 30   |
| <i>Bohadschia atra</i>       | 13                                                                                     | 2        | 4       | –                    | –                  | –    |
| <i>B. subrubra</i>           | –                                                                                      | –        | –       | 14                   | 8                  | –    |
| <i>Actinopyga mauritiana</i> | 5                                                                                      | –        | 9       | 3                    | –                  | 13   |
| <i>A. echinites</i>          | 3                                                                                      | –        | 6       | 16                   | 15                 | –    |
| <i>A. lecanora</i>           | 2                                                                                      | 2        | 2       | 10                   | 19                 | –    |
| <i>Stichopus hermani</i>     | 16                                                                                     | 1        | 9       | 17                   | 19                 | –    |
| <i>Thelenota ananas</i>      | 7                                                                                      | –        | 17      | 2                    | 4                  | –    |
| <i>T. anax</i>               | –                                                                                      | –        | 2       | –                    | –                  | 42   |
| Total                        | 100                                                                                    | 100      | 100     | 100                  | 100                | 100  |

that are frequently landed at Vanga are the two forms of *H. scabra* (*mchanga* and *barango*) which accounts for 43% of the total catch followed by *Stichopus hermani* that accounts for 16% of the catch (see Fig. 2). Catches at Majoreni are dominated by *H. scabra* (*mchanga*) that accounts for 95% of the catch. Shimoni was divided into three villages namely Shimoni, Kibuyuni and Kiromo. Sea cucumber catches at Shimoni are dominated by *H. scabra* (*mchanga*) that accounts for 20%, *Thelenota ananas* (*spinyo*) that accounts for 17% and *Holothuria nobilis* (*pauni mweusi*) that accounts for 17%. At Kibuyuni, *H. scabra* (*mchanga*) accounts for 22%, *S. hermani* (*tairi*) accounts for 17% and *Actinopyga echinites* (*mbura*) accounts for 16%. The catches at Kiromo are dominated by *H. scabra* (*mchanga*) (23%), *S. hermani* (*tairi*) (19%) and *Actinopyga lecanora* (*kitungu*) (19%). Catches at Gazi are dominated by *Thelenota anax* (*spinyo mama*) (42%) followed by *H. fuscogilva* (*pauni mweupe*) (30%).

### 3.3. Equipment used by fishers

The sea cucumber fishers use simple traditional equipment. In order to reach their fishing grounds, the fishers use traditional fishing vessels that consist of traditional canoes (*dau/mtumbwi*) that accounts for 57% of the sea cucumber fishing vessels (see plate 1), wooden planked boats (*mashua*) that account for 18%, motorized boats that account for 15%, and outrigger canoes (*ngalawa*) that account for 10%. From earlier studies e.g. Ochiewo [4], the type of fishing vessel was shown to determine catch because it influences how fast a sea cucumber fisher can reach the fishing ground and the amount of time available for actual fishing. It takes more time and energy to navigate a traditional canoe to a fishing ground thus a fisher that uses such a canoe normally remains with less time to fish compared to those fishers that use motorized or sail boats. Most of these fishing vessels are owned by entrepreneurs who rent them out to the fishers at a fee. Usually one vessel carries a number of fishers and when the fishers return from their fishing trips, each of them pays the vessel owner 20% of what he earns. Some vessel owners also act as first level middlemen who buy sea cucumbers from the fishers. Such vessel owners engage fishers who mainly collect sea cucumbers and sell to them. This practice ensures that buyers who own fishing vessels have a constant supply of sea cucumber irrespective of the price on offer. It has the disadvantage that it reduces the quantity of sea cucumbers exposed to competitive buying and suppresses the price of sea cucumber at the landing site since fishers are obliged to be loyal to the vessel owner or risk being barred from using the vessel again.



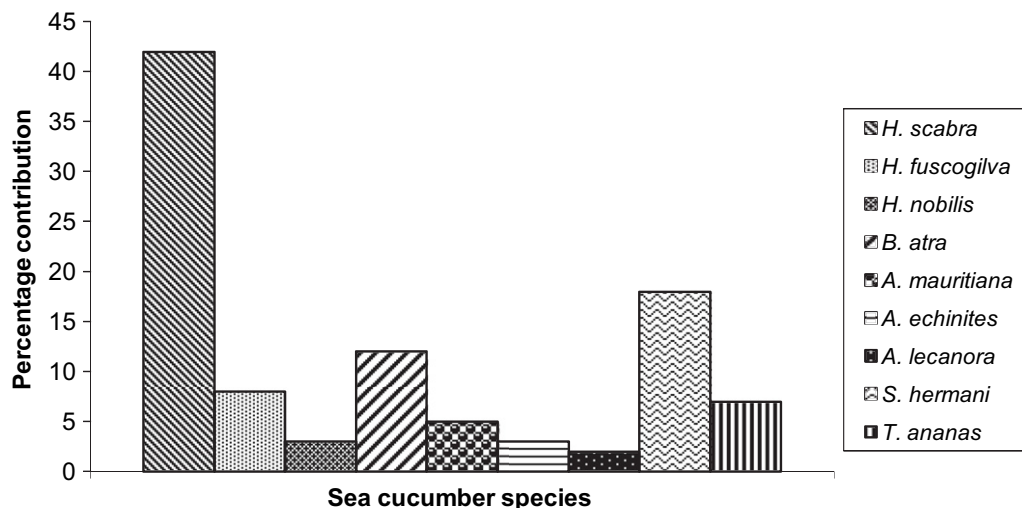


Fig. 2. Species composition of the sea cucumber catches at Vanga.

The other equipments which are used include snorkels, masks, fins and SCUBA to a limited extent. These equipments are used when a fisher is snorkeling or diving to collect sea cucumber from the bottom of the sea. Before the 1990s, snorkeling was the main method of sea cucumber fishing but this changed around 1992 when SCUBA diving was introduced [22]. SCUBA diving resulted in drastic increase in sea cucumber catch until an alarm was raised as people felt that the resource was being over-harvested and there was need to stop the use of SCUBA gear. At the moment, sea cucumber fishers in Kenya are depending on skin diving in the sub-tidal areas and snorkeling in the inter-tidal areas. Furthermore, a container (plate 2) or synthetic bags are also needed to store the

sea cucumbers while still in the sea. A container is necessary especially when a species of sea cucumber that has to be preserved in sea water has been caught. If this is not done then such species will decompose quickly and will have no value in the end. Some fishers however keep their catch in their fishing vessel without using a container or synthetic bag.

### 3.4. Fishing grounds

In most sites, sea cucumber fishing takes place in both inter-tidal and sub-tidal areas. It is only at Majoreni where sea cucumbers are mainly collected from the inter-tidal areas. The choice of fishing ground depends on the species being targeted and the skills required in the fishery. Some fishers are good at snorkeling and collecting sea cucumbers in the shallow inter-tidal waters while others have excellent SCUBA diving skills and are used to the deeper sub-tidal areas. Generally, it was noted that 48.3% of the sea cucumber fishers fish in sub-tidal areas while 46.7% fish in inter-tidal areas. At Vanga, sea cucumber fishing is mainly conducted in the sub-tidal area while at Majoreni it is mainly conducted in the inter-tidal area. About 5% of the sea cucumber fishers fish in both sub-tidal and inter-tidal areas. While most of the sea cucumber



Plate 1. A traditional sea cucumber fishing canoe at Majoreni.



Plate 2. A container used to store sea cucumbers while at sea.



**Plate 3.** A typical catch with *Holothuria scabra* dominating at Majoreni.

fishers operate in areas around their villages, there is a group of migrant sea cucumber fishers who move from one village to another. This was especially evident at Gazi where a group of migrant sea cucumber fishers from Shimoni occasionally go to camp and fish. During such a time, even a local middleman who is supposed to be based at Shimoni follows these sea cucumber fishers so that he could buy their catch immediately they land.

### 3.5. Status of the sea cucumber fishery

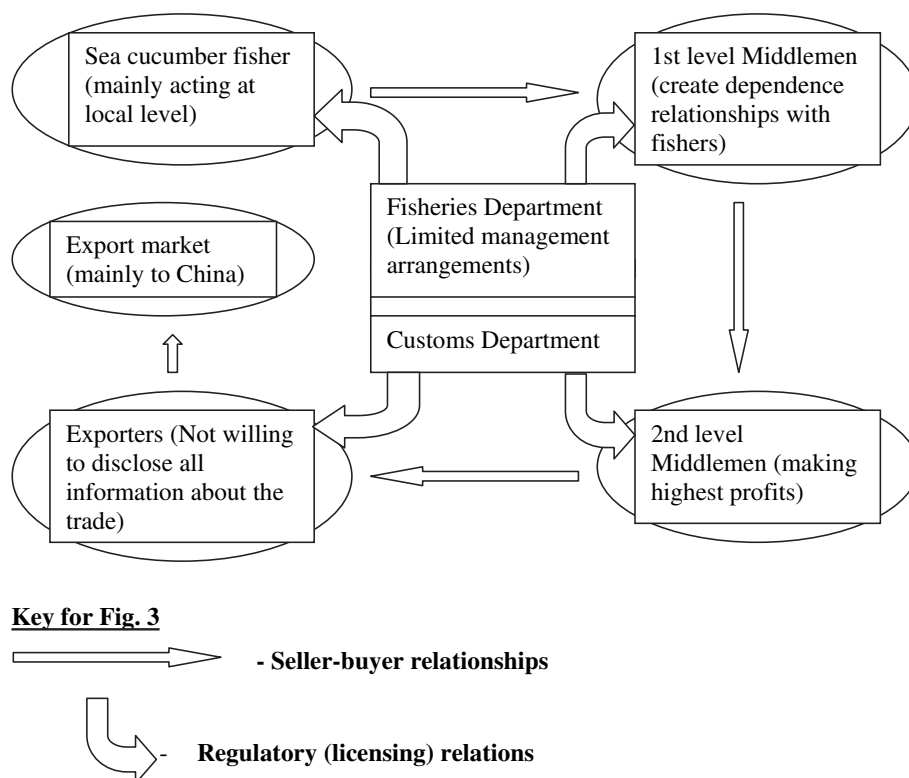
All the sea cucumber fishers prefer to land high commercial value species such as *H. fuscogilva*, *H. scabra* (both forms) (see plate 3) and *T. ananas*, but low-value species such as *S. hermani*, *Bohadschia atra*,

*A. echinites*, *Actinopyga mauritiana* and *Actinopyga lecanora* are increasingly being landed (see Table 5). These low-value species constitute over 40% of all the landed catch. This confirms the observation of Marshall et al. [21], and Beadle [7] that there is an increasing trend towards collection of the mid and low-value species such as *A. mauritiana*. Furthermore, the harvesting of sexually immature sea cucumbers is increasing [5] due to the high demand from the Asian market. The sea cucumber species are harvested from different habitats ranging from shallow reef areas, seagrasses to sandy bottoms. Most of the high value species are threatened by over-exploitation according to the fishers.

All the sixty-two fishers who were interviewed stated that sea cucumber catches have considerably decreased over the years. The decrease is both in terms of the quantity and size of individual sea cucumbers landed. Consequently, fishers are forced to spend more time fishing and land less sea cucumber than before. Those who have been collecting sea cucumbers for the last 20 years have stated that at present they land less than 50% of the quantity that they landed when they first entered the fishery. This decrease in sea cucumber stocks has also been reported by Beadle [7] who stated that *H. fuscogilva* and *H. nobilis* were increasingly difficult to find, and were only caught in small numbers using SCUBA equipment in Shimoni and Vanga. It is however important to note that the use of SCUBA was banned in 2003 after it was established that it led to over-harvesting of sea cucumbers. Furthermore, the low-value species are increasingly getting important to the fishers' catch to make up for the decrease in the size and quantities of high value species.

### 3.6. The sea cucumber market

The sea cucumbers are sold fresh from the sea to local middlemen who are referred in Fig. 3 below as first level middlemen. Here the sea cucumbers are sold in pieces and the



**Fig. 3.** Flow chart showing the links between the different actors involved in the sea cucumber trade i.e. fishers, middlemen, exporters and the different organizations in the Kenyan administration.

buyer decides on the price of each piece depending on the species and size landed. Since the sea cucumber is sold wet before being cleaned from the gut (stomach) contents, the buyer does not use a weighing balance at this stage but instead assesses the size using visual impression and estimating the size of each sea cucumber using the hand. For each species there is a minimum price which is paid to the fishers for the smallest size and a maximum price which is paid for the biggest size. The prices also vary from site to site. The price is largely controlled by the local level middlemen who dictate how much they pay for each piece of the species being sold. Each site has only one or two sea cucumber buyers (local level middlemen) hence there is virtually no competition for the sea cucumbers in the market. However, the price that is offered to the fishers who use fishing vessels provided by the middlemen is not so different from the price offered to fishers who use their own fishing vessels. The price that is paid to the sea cucumber fisher for the wet un-gutted sea cucumbers at the landing beach range from KSh.2–30 for low-value species such as *B. atra* and KSh.10–60 for *A. mauritiana*, *A. lecanora*, and *Holothuria fuscopunctata* to KSh.285 and KSh.475 (exchange rate is KSh.66 = US\$ 1) for high value species such as *H. scabra* ('barango') and *H. fuscogilva* respectively (see Table 6). The sea cucumber fishers occasionally borrow money from local middlemen especially when their catch is too small and they feel that the day's earnings are not sufficient for their household consumption needs. This credit facility in turn instills a loyalty towards middlemen thus creating potential conditions for dependence and exploitation.

Once the buyers (first level middlemen) purchase the sea cucumbers, they process them and sell to the second level middlemen at a profit. At this stage the sea cucumbers are weighed and sold in kilograms and the price depends on the species and size of sea cucumbers being sold. The prices range from KSh.120–450/kg for low-value species such as *Bohadschia subrubra* ('tambi mweupe') to KSh.200–800 for *A. echinites*, *A. lecanora*, and *Stichopus variegatus* (Table 6). However, there were significant differences in the price of processed sea cucumbers across villages. In some cases, the first level middlemen sell their processed sea cucumbers straight to the exporters who are also based in Mombasa.

It has been established that demand for sea cucumbers in the Asian market exceeds supply [7]. As the fishers try to meet this excess demand, over-exploitation has become evident and this has further resulted in lower catches and fishing of smaller (juvenile) sea cucumbers. In each landing beach there are first level middlemen who buy sea cucumbers from the fishers. For example, there are two middlemen at Vanga, seven at Shimoni and two at Majoreni. There are two exporters of sea cucumbers who are based in Mombasa. It has also emerged that sea cucumber exporters who are based in Zanzibar offer better prices compared to their counterparts in Kenya and this has attracted some Kenyan middlemen.

While sea cucumbers are highly valued by the local communities for the cash income that it generates, it is culturally not used as food item in the normal diet. No one has ever attempted to cook sea cucumbers. 95% of the respondents stated that they are not willing to taste sea cucumbers at all. The remaining 5% of the respondents stated that they are willing to taste it only if they are assured that it is edible and cannot harm them, and that they be shown how to process it to a point where it can be eaten. All the respondents stated that sea cucumbers are exported to China where they believe it is either a delicacy or a raw material for the manufacture of some other unknown products which may include medicine. Some of the reasons why people do not eat sea cucumbers in Kenya include the fact that sea cucumbers are culturally not food, sea cucumbers look fearsome, while some people simply said sea cucumbers just cannot be eaten.

The sea cucumber activity with the main actors, relationships and product flows is shown in Fig. 3.

### 3.7. Sea cucumber production and economic value

In this section we present the results of sea cucumber landings in three of the studied sites, i.e. Vanga, Shimoni and Majoreni and for Kwale district which combines all the studied villages. For Gazi Bay the data was not accurate as it was mixed with other villages and we decided to discard it. The data goes back to 2004 and thus represent the contemporary situation of the fishery and not the historical variation. It was also not possible to analyze the dynamics before and after the SCUBA introduction and ban. In general there is no trend towards a decrease or increase of sea cucumber landings during the period. Instead the fishery has been irregular and erratic with some better years than others. There are gaps in the data series, probably a consequence of restructuring of monitoring systems, improper recording, bad weather, annual leave for the data enumerators, etc. A few peaks are identified, e.g. at the end of 2007 and beginning of 2008. This peak may be due to higher effort and/or very good recruitment success in that season. However, another possible explanation is that catches coming from Northern Tanzania have been included in the counts. Especially for Vanga, cross-border issues are important and sometimes difficult to separate and control. The following figures show the landings for each site.

Calculations on economic value are extremely difficult to make, since prices between species vary substantially as well as the differences between wet and dry weights. The situation is the same for the rest of East Africa and in the global market. Since the sea cucumber landings presented above are for the whole range of mixed species, it is impossible to calculate the exact economic value of the landed product. In what follows we make some estimation exercises which represent the valuable species.

### 3.8. Estimated value and profit margin for *H. scabra*

We present here three estimations of the economic issues associated with *H. scabra* since this is the most dominant species in the area. In the global market sea cucumbers are economically target with a grade nominal scale in which grade 1 is the biggest size and best product, grade 2 is middle size and grade 3 is small size. We present two estimations for grades 1 and 3 and an estimation of the aggregated data for the villages of Vanga, Shimoni and Majoreni.

#### 3.8.1. Estimation 1. Big size sea cucumbers – high grade value (grade 1)

The price per piece of grade 1 *H. scabra* (wet weight) is Ksh.200 (US\$ 2.8).<sup>2</sup> One kilogram requires six to eight pieces, so the earnings per fisherman per kilogram range from between Ksh.1200 (US\$ 16) and Ksh.1600 (US\$ 21), depending on the number of pieces. This implies that the buying price is in the range of Ksh.1200–1600/kg. However, the price offered to the dealers/middlemen by the exporters in Mombasa is Ksh.2100 (US\$ 28)/kg, hence the profits from the product are Ksh.900 (US\$ 12) or Ksh.500 (US\$ 6.7)/kg. This indicates that the middleman profit ranges from 31% to 75% per kilogram of dry grade 1 *H. scabra*. The high profits generated may provide an incentive to the middleman to press the fisherman and keep on earning high profits.

The next profit level is from the middleman to the exporters based in Zanzibar or Tanga in Tanzania. This analysis assumes that

<sup>2</sup> The exchange rate is US\$ 1 = Ksh.75.

**Table 6**

Prices of sea cucumbers in the Kenyan market, with local names given (in brackets). The prices vary depending on whether the product is fresh (wet weight) or it has been dried (dry-weight).

| Species                               | Wet weight prices (KSh.) |         | Dry weight prices (KSh.) offered by exporters in Kenya |         | Dry weight prices (KSh.) offered by exporters from Tanzania |
|---------------------------------------|--------------------------|---------|--------------------------------------------------------|---------|-------------------------------------------------------------|
|                                       | Minimum                  | Maximum | Minimum                                                | Maximum |                                                             |
| <i>H. scabra</i> ('mchanga')          | 11                       | 200     | 620                                                    | 2100    | –                                                           |
| <i>H. scabra</i> ('barango')          | 30                       | 285     | 1000                                                   | 1500    | 2885                                                        |
| <i>H. fuscogilva</i> ('pauni mweupe') | 150                      | 475     | 900                                                    | 2800    | 4040                                                        |
| <i>H. nobilis</i> ('pauni mweusi')    | 40                       | 200     | 400                                                    | 800     | 1440                                                        |
| <i>H. fuscopunctata</i> ('sumu')      | 10                       | 30      | –                                                      | –       | –                                                           |
| <i>B. atra</i> ('tambi mweusi')       | 2                        | 30      | –                                                      | –       | 400                                                         |
| <i>B. subrubra</i> ('tambi mweupe')   | –                        | 10      | 120                                                    | 150     | 400                                                         |
| <i>A. echinites</i> ('mbura')         | 2                        | 30      | 300                                                    | 600     | 808                                                         |
| <i>A. lecanora</i> ('kitungu')        | 5                        | 20      | 200                                                    | 400     | –                                                           |
| <i>A. mauritiana</i> ('kijino')       | 10                       | 60      | –                                                      | –       | –                                                           |
| <i>T. anax</i> ('spinyo mama')        | 20                       | 30      | 150                                                    | 170     | –                                                           |
| <i>T. ananas</i> ('spinyo')           | 60                       | 250     | 600                                                    | 2000    | 2020                                                        |
| <i>S. variegates</i> ('tairi')        | 5                        | 65      | 200                                                    | 800     | 1765                                                        |

**Table 7**

Basic economic data for the landings of three villages in the study, i.e. Vanga, Shimoni and Majoreni. Total landings are the sum of all reported sea cucumbers over the period (2004, 2005–2008). The average and standard deviation are per recorded data per month. The minimum and maximum data corresponds to all reported data per site during the given period. The biomass data (kg) is the actual value recorded by KMFRI and Dept. of fisheries in coordination. However, the economic value is a calculated estimation based on the assumptions of having a dominating catch of *Holothuria scabra* and an average price per kg of US\$ 10.6. See text for estimation details.

| Site     | Total landings |            | Average |            | Standard deviation |            | Min–max |            |
|----------|----------------|------------|---------|------------|--------------------|------------|---------|------------|
|          | kg             | Value (US) | kg      | Value (US) | kg                 | Value (US) | kg      | Value (US) |
| Vanga    | 14,972.0       | 158,703.2  | 356.4   | 3778.6     | 112.9              | 1197.2     | 150–700 | 1590–7420  |
| Shimoni  | 13,021.0       | 138,022.6  | 351.9   | 3730.3     | 212.6              | 2253.8     | 55–1120 | 583–11,872 |
| Majoreni | 1605.0         | 17,013.0   | 43.3    | 459.8      | 70.5               | 747.78     | 12–450  | 127–4770   |

the cost of transport is constant since the exporters from Tanzania often make their own transportation arrangements. The price of the product at this chain level is Ksh.3000 (US\$ 40)/kg of *H. scabra* with a gross profit to the middleman, when the product is sold to exporters, ranging from Ksh.1400 (US\$ 18.7) to Ksh.1800 (US\$ 24). The gross profit from the initial chain level to the top level is between 88% and 150% per kilogram.

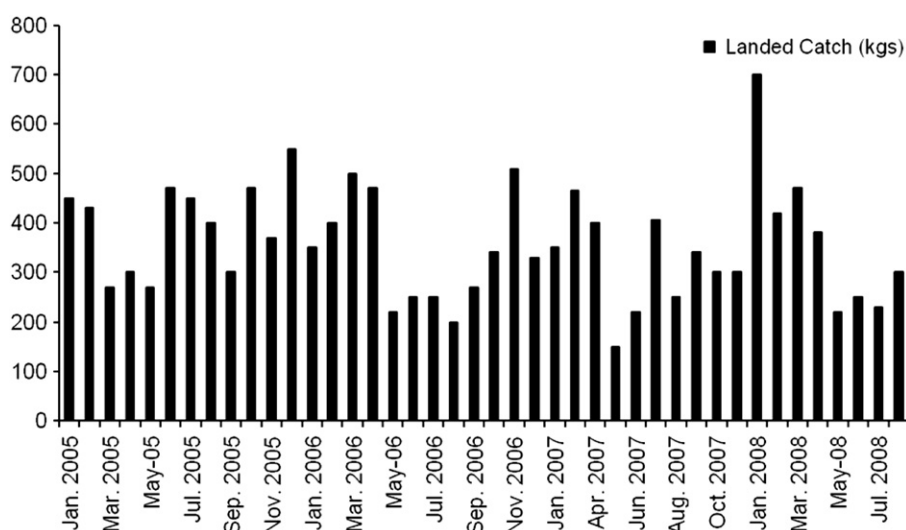
### 3.8.2. Estimation 2. Small size – low grade value (grade 3)

In this case, about 12 pieces are needed for a kilogram of the product, and the price per piece (wet weight) is Ksh.12. The

earnings per fisherman are then about Ksh.144/kg. The market price offered in the next chain step is Ksh.620/kg. That means that the earned profit is Ksh.476/kg. The profit at the top level of the chain in this case is as high as 330% for the low grade *H. scabra*. This is thus also a clear incentive for the middleman to encourage fishermen to land juvenile sizes of the valuable species.

### 3.8.3. Estimation 3. Value ranges of the sea cucumber landings at the different sites

As stated before, it is not possible to provide the exact landing value of the sea cucumbers because all species are mixed and sizes



**Fig. 4.** Monthly landings of sea cucumbers in Vanga village, Kenya. Data shows dry weight in kilograms during the period of 2005–2008.



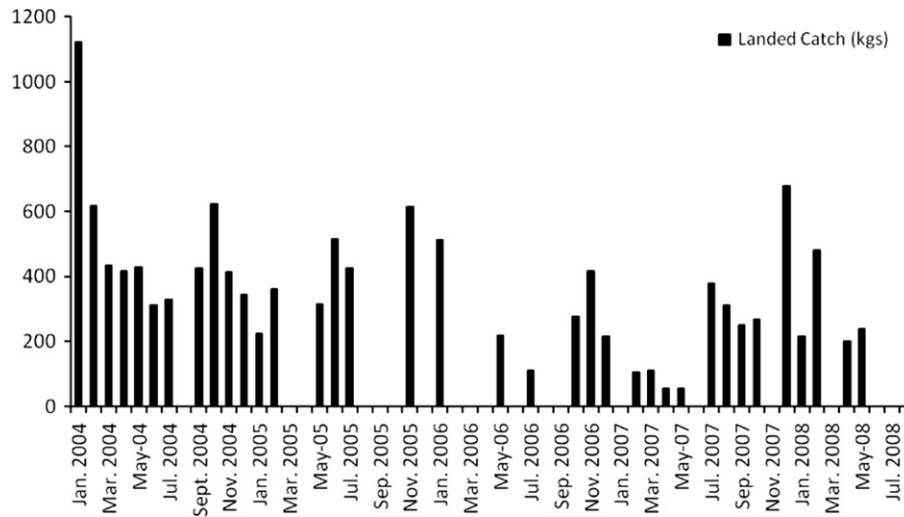


Fig. 5. Monthly landings of sea cucumbers in Shimoni village, Kenya. Data shows dry weight in kilograms during the period of 2004–2008.

unknown. But to get an idea on the order of magnitude of the total catch value during 2004(5)–2008, a mean value for *H. scabra* was assumed to estimate the total catch value. As an estimate this figure should be interpreted with caution. An average kilogram value, based on the prices given to fishermen of big and small sea cucumbers was calculated at Ksh.772/kg, rounding up to Ksh.800 (or US\$ 10.6)/kg.

Table 7 presents the general characteristics of the landings in the villages where statistics of good quality were found, i.e. Vanga, Shimoni and Majoreni. The results show that Vanga and Shimoni sites are similar in total landings, however the standard deviation for Shimoni is larger. Majoreni production is very small compared to the other two sites, and may reflect ecological differences, under-recording of catches since the main sea cucumber landing beaches are far apart and are different from where finfish are landed, or social aspects since in this village the sea cucumber fishery is taking place as a response to overfishing of finfish fisheries (see study site descriptions). The difference between the minimum and maximum production values was large for all sites due to marked variations in catches (Figs. 4–7). However, the average economic value per month for the Vanga and Shimoni sites can be substantially high, of the order of US\$ 3778 and 3730 respectively.

### 3.9. Supplementary livelihood activities

While 68% of the respondents depended on sea cucumber fishing as their main source of income especially during the

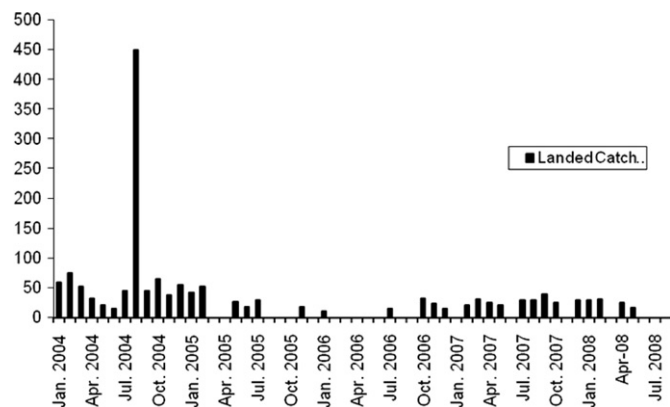


Fig. 6. Monthly landings of sea cucumbers in Majoreni village, Kenya. Data shows dry weight in kilograms during the period of 2004–2008.

northeast monsoon, they also engage partly in peasant farming, octopus and finfish fishing, and small informal businesses to supplement their income. 32% of the respondents depended on finfish and octopus fishing as their main source of income with sea cucumber fishing playing a supplemental role. Peasant farming and casual jobs are mainly important during the southeast monsoon when the water is turbid thus impairing visibility. The southeast monsoon also coincides with the end of the long rains that support peasant farming activities.

### 3.10. Management of the sea cucumber fishery

Sea cucumber fishery is regulated by the Fisheries Act cap 378 of the Laws of Kenya [19]. The Fisheries Department is charged with the responsibility of enforcing this legislation. The existing management strategies include licensing of fishers and fishing vessels, licensing of fish traders including middlemen and exporters in the sea cucumber fishery, collection of catch statistics, and regulation of fishing techniques. According to the Fisheries Act cap 378, fishers pay Ksh.100 for the fishing license annually. Furthermore, the exporters pay a *beche-de-mer* export fee which is calculated at 5% *ad valorem* of the market price. In order to collect this export fee, the exporters are required to declare the quantities and values of sea cucumbers that are exported at a particular time. There is however a tendency for exporters to declare less than the actual values exported. More recently, the involvement of local communities in fisheries management has been embraced through the establishment and strengthening of Beach Management Units (BMUs). The BMUs refer to community based organizations that have been established by the Fisheries Department to give the local communities power to participate in the management of fisheries and related aquatic resources. The Beach Management Units were first established and gazetted in 2007 to give them legal backing. They have control over the activities that take place in their fishing grounds and fish landing beaches. Since their establishment, most of the BMUs are successfully operating and the Fisheries Department is providing support to them in terms of training to enable them to understand their roles effectively and to build their capacity in management. The BMUs would therefore provide a mechanism for the sea cucumber fishers to participate in managing the fishery.

Apart from the main fisheries legislation, there is no special law or policy that is devoted specifically to manage and regulate the sea

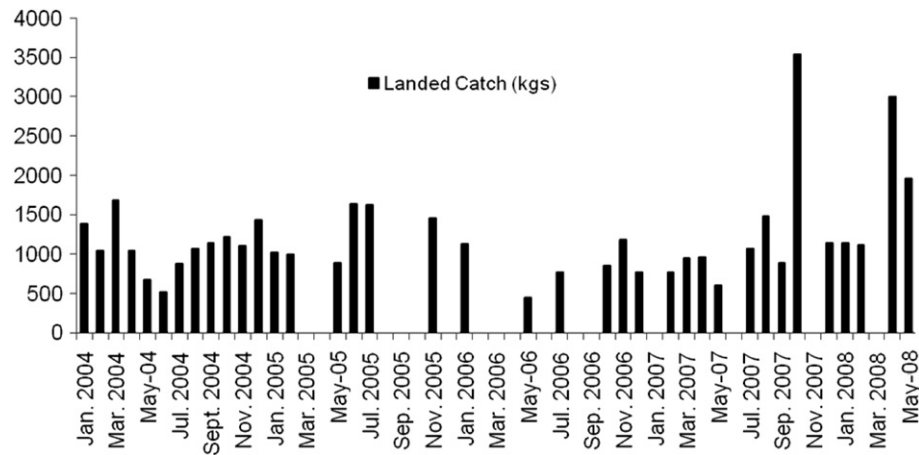


Fig. 7. Monthly landings of sea cucumbers in the former Kwale district, where the villages are located. Data shows dry weight in kilograms during the period of 2004–2008.

cucumber fishery in the area. About 50% of the interviewees pointed out that so far there is no regulation on the size of individual sea cucumbers harvested. This is an incentive for fishers to harvest what they feel among themselves is appropriate. Consequently, the fishing of small-size sea cucumbers is increasing and is likely to become a problem in future, especially if recruitment relies on local stocks.

Kenya has established MPAs as a management tool for the marine and coastal resources as provided for in the Wildlife Conservation and Management Act (1989) [20]. So far four marine parks have been established and are being managed as no take areas. All biological resources within a marine park are protected and no removal or disturbance of any living or non-living marine resource is permitted except if necessary for monitoring or research to evaluate management effectiveness. Six marine reserves have also been established where only traditional fishing methods are allowed [4,16–18]. The central objective for the establishment of MPAs in Kenya, and the rest of the Eastern African region, is to conserve and manage the extraordinary variety of the marine and coastal biodiversity for the benefit of the present and future generations [16]. The sea cucumber stocks, like other biological resources in the marine parks, are managed by the Kenya Wildlife Service while stocks in the marine reserves fall under the jurisdiction of both Kenya Wildlife Service and Fisheries Department.

The situation calls for urgent management attention. Regulation, monitoring, enforcement and revision are some of the first steps to improve the management situation. However, since sea cucumber constitutes an important livelihood, stressed by the facts that 68% of those engaged in sea cucumber fishing are full-time fishers, a majority are heads of the households and married. Thus management efforts should integrate the livelihoods component and the links between the resource users and the environment. In this sense, the BMUs may provide an opportunity and access point from which to plan and promote management schemes together with the resource users.

From an economic point of view, the sea cucumber production provides good revenues as shown in Table 7. Although the estimation of the value lies on the higher end since we consider a high value catch of *H. scabra*, the revenues can be substantially high even with a mixed catch. Rounding the values from Table 7, we found a monthly average value for the studied area to be US\$ 8000. However, the calculations on profit from the different sea cucumber sizes have shown that collecting the smaller sizes provides the highest relative profits. The identification of this incentive is a key to promotion of sustainability of the activity, since removing smaller size individuals may jeopardize recruitment, eventually

leading to a fishery collapse. In addition, there is very little information in the Western Indian Ocean region of larval dynamics and population connectivity. The stocks may be dependent on local or regional stocks. Considering this important knowledge gap, a precautionary approach is truly necessary. The latter, combined with the increased fishing effort (shown in Table 4), can potentially be devastating for the fishery.

The lack of targeted management and the few regulations seems to be insufficient to manage this valuable resource. A comprehensive revision is needed integrating sea cucumber management in the broader scheme of fisheries management.

The study also suggests that the commerce network connections may be highly complex, with commercial connections at local, regional and international levels. There are connections between villages, for example the case of Shimoni and Gazi where the commodity moves at local spatial scales. There are also connections between countries of the WIO. Important links were identified between Kenya and Tanzania (mainly through Zanzibar, but also cross-borders in the mainland). The commodity moves further into the international market and the study suggests that China is an important market. Such multiple scale connections call for management actions at different levels, though it is too early to define how these management arrangements may be structured. Suffice to say that it is important to keep these issues in mind for the future.

#### 4. Conclusions

The results show that *H. scabra* is dominant in the sea cucumber catch while the catches of the other high commercial value species, *H. fuscogilva*, have declined over the years, a sign of over-harvesting. This species may be threatened with depletion if control measures are not put in place urgently. Collection of the previously low-value species and small-size sea cucumbers is increasing due to high demand in the Asian market and the high profits from these sizes. There is demand for different grades of processed sea cucumbers, grade 1 (the biggest size sea cucumbers), grade 2 (the medium size sea cucumbers) and grade 3 (the small-size sea cucumbers). Grade 3 sea cucumbers include low-value species and juvenile sea cucumber catches. The demand for grade 3 sea cucumbers may result into increased fishing of juveniles and may consequently affect stock reproduction and recruitment. It may therefore be important to establish size limits for sea cucumbers that enter the market, potentially enforced by checking the length and width of processed sea cucumbers.

There are strong indications that the fishers are exploited by local level middlemen since they often do not have a say over the prices given to them and the time to process their catch themselves. For a few fishers who have tried to process their catch, they do not have the needed training to enable them to meet the high quality standards and they are not in contact with the exporters who are based in Mombasa. It is therefore important to provide training to the fishers on sea cucumber processing so that those who are ready to process their catch can do so. It will also be useful for the sea cucumber fishers to form an association that can market their catch. The association could negotiate competitive prices with the buyers and undertake sea cucumber processing on behalf of the members when necessary. If these were done, the fisherman dependence relationship with the middlemen will be changed.

In order to build a reliable database and establish trends in the sea cucumber fishery, there is a need to make it mandatory for licensed owners of sea cucumber fishing vessels to provide data on catch and effort, fishing ground where sea cucumbers have been caught and the number of fishers who have used the boat. Further, a special arrangement should be made with the sea cucumber exporters to provide accurate data on the quantity of each species exported, the duration taken to accumulate the quantities, the destination of his consignment, and the price offered for each species. The information on price offered in each export market could then be treated as confidential and only be used for analysis purposes and to encourage investors who may be interested in sea cucumber mariculture.

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